Decompilation Tools – APKTool aur JADX

Reverse engineering mein do sab se important tools APKTool aur JADX hain. Dono ka kaam alag hai aur dono ek doosre ke complementary hain. Ek saath use karne se aapko APK ka complete analysis milta hai.

APKTool – Resource aur Smali Level Tool

APKTool APK ko low-level par decompile karta hai. Iska output smali code (Dalvik bytecode ki human-readable form), resources, aur AndroidManifest.xml hota hai.

Install Kaise Karein

macOS (Homebrew):

```
brew install apktool
```

Linux (apt):

```
sudo apt update
sudo apt install apktool
```

Windows: Official website se `apktool.jar` aur `apktool.bat` download karein, PATH mein add karein.

Basic Usage

APK ko decode karna (resources + smali):

```
apktool d app.apk -o output_folder
```

`s` flag se dex files ko skip kar sakte hain (faster, sirf resources chahiye):

```
apktool d -s app.apk -o output_folder
```

Modifications ke baad APK rebuild karna:

```
apktool b output_folder -o modified_app.apk
```

APKTool Kya Deti Hai?

- Smali code – Low-level assembly jaisa code, jo logic modify karne ke kaam aata hai
- res/ folder – Layouts, images, aur saari resources
- AndroidManifest.xml – Decoded XML format mein
- assets/ folder – Raw files (fonts, audio, etc.)

APKTool Kaise Kaam Karti Hai?

APKTool resources ko decode karti hai aur smali code generate karti hai. Yehi woh tool hai jo aapko APK ko modify karne, rebuild karne, aur wapas sign karne ki facility deti hai. Apktool `AndroidManifest.xml` se `<uses-sdk>` element (minSdkVersion, targetSdkVersion) ko hata kar `apktool.yml` file mein daal deti hai. Ye normal behaviour hai.

JADX – Java Source Code Decompiler

JADX APK ke `classes.dex` files ko readable Java/Kotlin source code mein decompile karta hai. Code logic ko samajhna, API calls track karna, aur vulnerabilities find karna ismein bohot easy hai.

Install Kaise Karein

macOS (Homebrew):

```
brew install jadx
```

Linux (apt):

```
sudo apt install jadx
```

GUI Version: Official GitHub releases se zip download karein aur `jadx-gui` executable run karein.

Basic Usage

Command Line:

```
jadx -d output_folder app.apk
```

GUI Version (Recommended):

```
jadx-gui app.apk
```

GUI mein left panel mein package structure, right panel mein decompiled Java code dikhta hai. Search feature (Ctrl+Shift+F) se strings, classes, ya methods quickly find kar sakte hain.

JADX Kya Deti Hai?

- Java/Kotlin source code – Readable, high-level code
- Decompiled AndroidManifest.xml – `--no-src` flag se sirf resources extract kar sakte hain
- Search aur Navigation – Codebase mein quick search aur navigation

JADX Kaise Kaam Karti Hai?

JADX APK ke `classes.dex` files ko analyze karti hai aur unhein Java bytecode se readable source code mein convert karti hai. Obfuscated code bhi partially readable ho jata hai. Agar APK Kotlin mein likhi hai toh decompiled code Java syntax mein dikhega, lekin logic bilkul original jaisa hi rahega.

APKTool vs JADX – Key Differences

Output: APKTool smali code aur resources deta hai jabke JADX Java/Kotlin source code deta hai.

Readability: APKTool ka output low-level aur assembly jaisa hota hai jabke JADX ka output high-level aur readable hota hai.

Modification: APKTool mein smali aur resources modify kar sakte hain jabke JADX sirf analysis ke liye hai.

Rebuild: APKTool APK rebuild kar sakti hai jabke JADX yeh facility nahi deta.

Use Case: APKTool resource analysis aur smali patching ke liye use hoti hai jabke JADX code logic analysis, API calls tracking, aur vulnerability hunting ke liye use hoti hai.

Kab Kaunsa Tool Use Karein?

JADX use karein jab:

- Code logic samajhna ho
- API calls, authentication flow, sensitive data handling analyze karna ho
- Hardcoded credentials ya keys find karni hon
- App ka architecture aur flow samajhna ho
- Quick code review karna ho

APKTool use karein jab:

- XML resources modify karni hon (strings, layouts, colors, etc.)
- AndroidManifest.xml change karna ho
- Smali code patch karke app behavior change karna ho
- APK rebuild karke repack karna ho

Workflow – APK Analysis Kaise Karein

Step 1: JADX se decompile karein

```
jadx-gui app.apk
```

Code analyze karein, interesting strings aur classes search karein.

Step 2: APKTool se decompile karein

```
apktool d app.apk -o decompiled/
```

Step 3: Resources modify karein (agar zaroorat ho)

- strings.xml mein text change karein
- layout XMLs modify karein
- AndroidManifest.xml permissions change karein

Step 4: Smali code modify karein (agar logic change karna ho)

- JADX se logic samjhein
- APKTool ke smali code mein uss logic ko patch karein

Step 5: APK rebuild karein

```
apktool b decompiled/ -o modified.apk
```

Step 6: APK sign karein

```
jarsigner -verbose -sigalg SHA1withRSA -digestalg SHA1 -keystore my.keystore modified.apk alias
```

📘 **Educational Purpose Only:** Yeh guide sirf educational aur learning purposes ke liye hai. APK decompilation ka istemal sirf apne own apps aur authorized environments mein karein. Kisi doosre ke app ko bina permission ke modify ya crack karna applicable cyber laws ke under punishable hai.

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